

Fuel jettison

FQMS fuel jettison with the JTSN GW, the ECAM memo and procedures

Branch	feature/a380/fuel-jettison
Aircraft	A380X
Commits	b04a3851d feat(a380x): fuel jettison with JTSN GW, ECAM memo and procedures
Base	master 280800cf7
Status	Merged in master. Node-simulated; deployed; in-sim test pending.

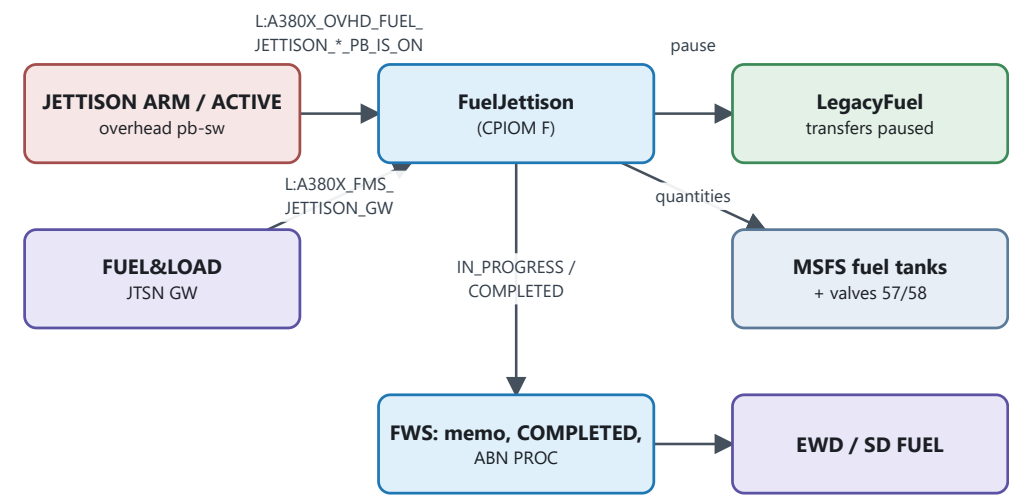
Purpose

FBW had the overhead JETTISON ARM / ACTIVE pushbuttons only (they toggled LVars). The JTSN GW field of FUEL&LOAD was disabled.

Specification (A380 FCOM DSC-28-40 Fuel jettison)

- CPIOM F `FuelJettison`: with ARM and ACTIVE on, fuel leaves the transfer tanks (proportionally) at the FCOM rate; the feed tanks are never jettisoned.
- Trim tank: held at the forward T.O/LDG CG limit (from the airframe data) and resumed at +0.5 %.
- Stops at the JTSN GW, when the transfer tanks are empty, or when a pb-sw is set OFF; COMPLETED latches until both pb-sw are OFF. The other fuel transfers pause during the jettison.
- The MSFS fuel system has no overboard outlet: the tank quantities are set directly (valves 57/58 open for the SD).
- FUEL&LOAD JTSN GW entry: from ZFW + 11 000 lb to the maximum GW (FCOM DSC-22-FMS-20-100), sent to the FQMS.
- ECAM: memo JETTISON IN PROGRESS; sensed FUEL JETTISON COMPLETED (ACTIVE OFF, ARM OFF, sensed); ABN PROC FUEL JETTISON (FUEL menu) with its FWS entry, the ARM / ACTIVE lines sensed. Cockpit: the ACTIVE ON light goes off when completed.

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

Interfaces

LVar	Written by	Meaning
------	------------	---------

L:A380X_FMS_JETTISON_GW	FMC	JTSN GW in kg
L:A380X_FUEL_JETTISON_IN_PROGRESS	FuelJettison	jettison running
L:A380X_FUEL_JETTISON_COMPLETED	FuelJettison	stopped by the FQMS, until both pb-sw are OFF

Limitations

- FAULT, VLV NOT CLOSED and INOP SYS are not wired (FBW has no jettison valve failures).
- The behaviour XML change (ACTIVE light) needs a full MSFS restart.

PR description (from the PR draft)

From `pr-drafts/15-fuel-jettison.md`, section "Summary of Changes" onwards.

Summary of Changes

FBW had the overhead JETTISON ARM / ACTIVE pb-sw only. Fuel jettison per the A380 FCOM DSC-28-40:

- **FQMS (CPIOM F FuelJettison)**: with ARM and ACTIVE on, fuel leaves the transfer tanks (proportionally) at the FCOM rate; the feed tanks are never jettisoned. The trim tank is held at the forward T.O/LDG CG limit and resumed at +0.5 %. The jettison stops at the JTSN GW, when the transfer tanks are empty or when a pb-sw is set OFF; COMPLETED latches until both are OFF. The other transfers pause meanwhile. The MSFS fuel system has no overboard outlet, so the tank quantities are set directly (valves 57 / 58 open for the SD).
- **FUEL&LOAD**: the JTSN GW field, from ZFW + 11 000 lb to the maximum GW (DSC-22-FMS-20-100), sent to the FQMS.
- **ECAM**: memo JETTISON IN PROGRESS; FUEL JETTISON COMPLETED (ACTIVE OFF, ARM OFF, sensed); ABN PROC FUEL JETTISON (FUEL menu) with its FWS entry, the ARM / ACTIVE lines sensed.
- **Cockpit**: the ACTIVE ON light goes off when the FQMS completes the jettison.

Cockpit API Changes

New LVars: `L:A380X_FMS_JETTISON_GW` (kg, FMC), `L:A380X_FUEL_JETTISON_IN_PROGRESS`, `L:A380X_FUEL_JETTISON_COMPLETED` (FQMS). Reads `L:A380X_OVHD_FUEL_JETTISON_ARM_PB_IS_ON` / `_ACTIVE_PB_IS_ON`.

Screenshots (if necessary)

TO ADD: FUEL&LOAD JTSN GW, the SD FUEL page during a jettison, the memo, the COMPLETED procedure, the ABN PROC procedure.

References

- A380 FCOM DSC-28-40 (fuel jettison), DSC-22-FMS-20-30 / -20-100 (FUEL&LOAD JTSN GW), PRO-ABN-ECAM-10-28 / -20-28 (procedures).

Additional context

FAULT, VLV NOT CLOSED and INOP SYS are not modelled (FBW has no jettison valve failures). The behaviour XML change needs a full MSFS restart.

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. In flight, JTSN GW = GW - 20 t, ARM then ACTIVE: memo, fuel decreasing on the SD, trim tank held at the CG limit.
2. At the JTSN GW: jettison stops, COMPLETED procedure, ACTIVE light off; both pb-sw OFF clears it.
3. ABN PROC > FUEL > FUEL JETTISON: the procedure shows, ARM / ACTIVE lines tick with the pb-sw.